



COMSATS University Islamabad

Department of Computer Science

Course Description Form (CDF)

Course Information

Course Code: **CSC211**

Course Title: **Data Structures**

Credit Hours: **4(3,1)**

Lecture Hours/Week: **3**

Lab Hours/Week: **3**

Pre-Requisites: **CSC103-Programming Fundamentals**

Course Objective

- To discuss the issues of time complexity and examine various algorithms from this perspective;
- To introduce the concept and usage of data structures through abstract data structures, including linked lists, stacks, queues, priority queue, trees, and graphs;
- To implement above data structures and their applications;
- To develop an understanding of recursion as they apply to trees and graphs.

Course Content

This course provides fundamental knowledge of data organization. The topics include: Overview of Data Structures; Static & Dynamic List; Stack; Queue; Tree & its Algorithms; Graph & its Algorithms; Sorting; Searching; Hashing; and Time Complexity of an Algorithm.

Unit wise Major Topics:

Unit	Topics	No. of Teaching Hours
1.	Data Structure: Overview, Importance, Classification, Operations, and Abstract Data Types.	1.5
2.	Static List; Dynamic List: Single Linked Lists, Circular Linked List, Doubly Linked List, and Applications.	9
3.	Stack: Concept, and Applications.	3
4.	Queue: Concept, Types and Applications.	4.5
5.	Tree: Concept & Terminology, Traversal Algorithms; Binary Search Trees & its Various Operations; AVL Tree and Min & Max Heaps.	9
6.	Graph: Basic Concepts & Terminology; Representation, Types; Graph Traversal Techniques: Breadth First Search, Depth First Search; Minimum Spanning Trees: Kruskal Algorithm, Prims Algorithm; and Shortest Path Problem: Dijkstra's Algorithm.	7.5
7.	Sorting algorithms: Bubble, Selection, Insertion & Merge Sort, Searching Algorithms: Linear & Binary Search; Hashing: Hash Functions, Hash Tables, Strategies for Avoiding & Resolving Collisions.	6
8.	Complexity Analysis: Growth Rate of Function; Asymptotic Notation; Time Complexity of Searching and Sorting Algorithms.	4.5
Total Contact Hours		45

Mapping of CLOs and GAs

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy	GA
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			Learning Level		
CLO's for Theory					
CLO-1	1-4	Employ linear data structures to solve computing problems.	Applying	2	
CLO-2	5-6	Use non-linear data structures to solve computing problems.	Applying	2	
CLO-3	7-8	Analyze the time complexity of various algorithms.	Analyzing	2,3	
CLO's for Lab					
CLO-4	1-7	Implement data structures and algorithms.	Applying	2,4	
CLO-5	1-8	Develop a project using appropriate data structures in a team environment.	Creating	2-4, 6	
CLO Assessment Mechanism					
Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5
Quizzes	Quiz 1&2	Quiz 3	Quiz 4	-	-
Assignments	Assignment 1&2	Assignment 3	Assignment 4	Lab Assignments	-
Mid Term Exam	Mid Term Exam	Mid Term Exam	-	Lab Mid Term Exam	-
Final Term Exam	Final Term Exam			-	Lab Project/ Lab Final Term Exam
Text and Reference Books					
Textbook:					
1. A Common-Sense Guide to Data Structures and Algorithms, Jay Wengrow, Pragmatic Bookshelf, 2020.					
Reference Book:					
1. Data Structures and Algorithm Analysis in C++, Mark Allen Weiss, Addison-Wesley, 2014.					